

5J Index laws for multiplication and division

Learning Objective	To be able to evaluate and simplify exponent forms through the use of index laws.
Today We Are	Multiplying and dividing numbers in index forms.
Indicators of Progress	• I can multiply and divide numbers in index form

Recall Activity

Expand the following:



$$6a (2 + 3a)$$

Factorise the following:

$$4a + 6ab + 8ac$$

Lets have a look at Index Forms:

- 3 squared = $3 \times 3 = 3^2$
- 3 cubed = $3 \times 3 \times 3 = 3^3$

What would the index form look like if I had five 3's?

$$3 \times 3 \times 3 \times 3 \times 3 =$$

➡ Index / Exponent / Power

Base 5_6



X



X



X



X

$$4^{2+3} = 4^5$$

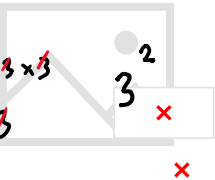


X

$$= 2^{2+3+4+1}$$

$$= 2^{10}$$

$$= 2$$

$$\begin{array}{c}
 3 \times 3 \times \cancel{3} \times \cancel{3} \times 3 \\
 3 \times \cancel{3} \times \cancel{3} \times 3
 \end{array}$$


$$3^{5-3} = 3^2$$



$$7^{6-3} \neq 7^3$$

$$6^{8-5} = 6^3$$

Fluency





g h i j

Problem Solving

